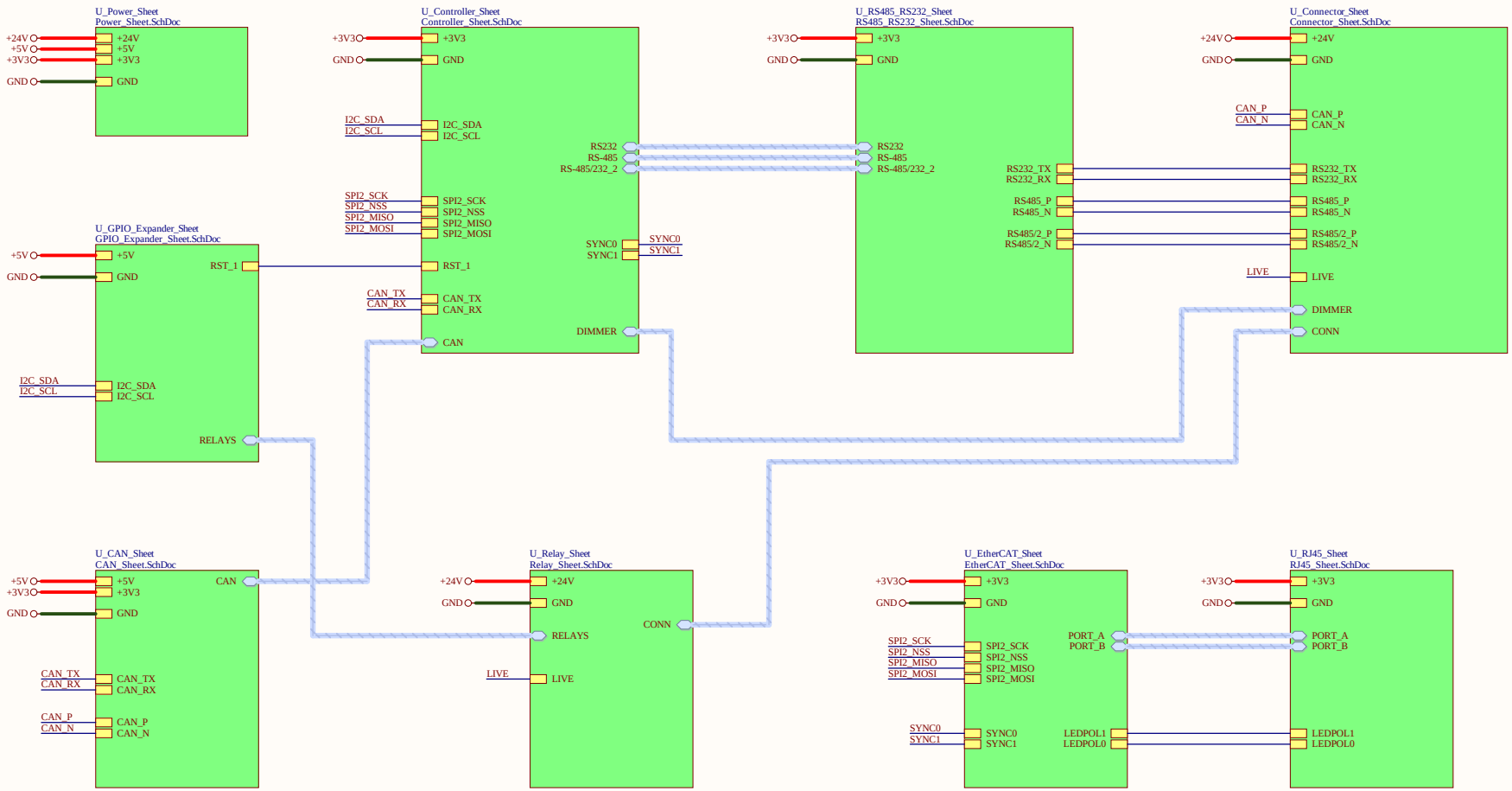
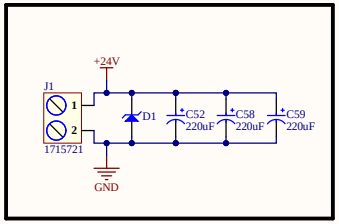


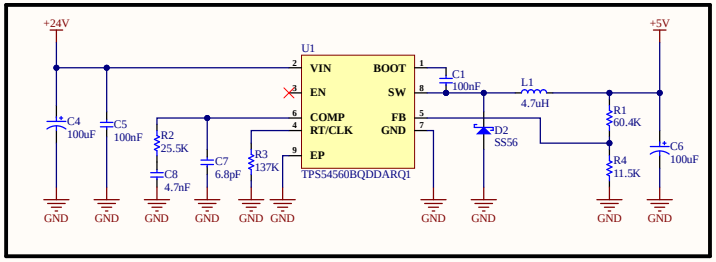
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Drawn By: Huzaifa Ali	Page Total: 10

24V INPUT

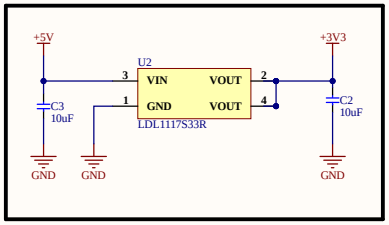


5V 5A BUCK



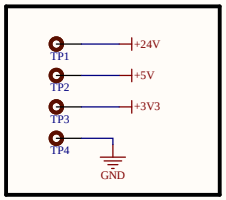
5A

3.3V 1A LDO

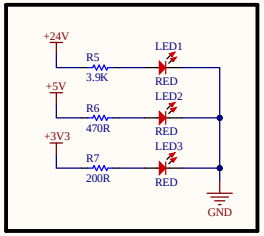


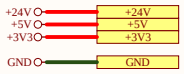
1.2A

TEST POINTS



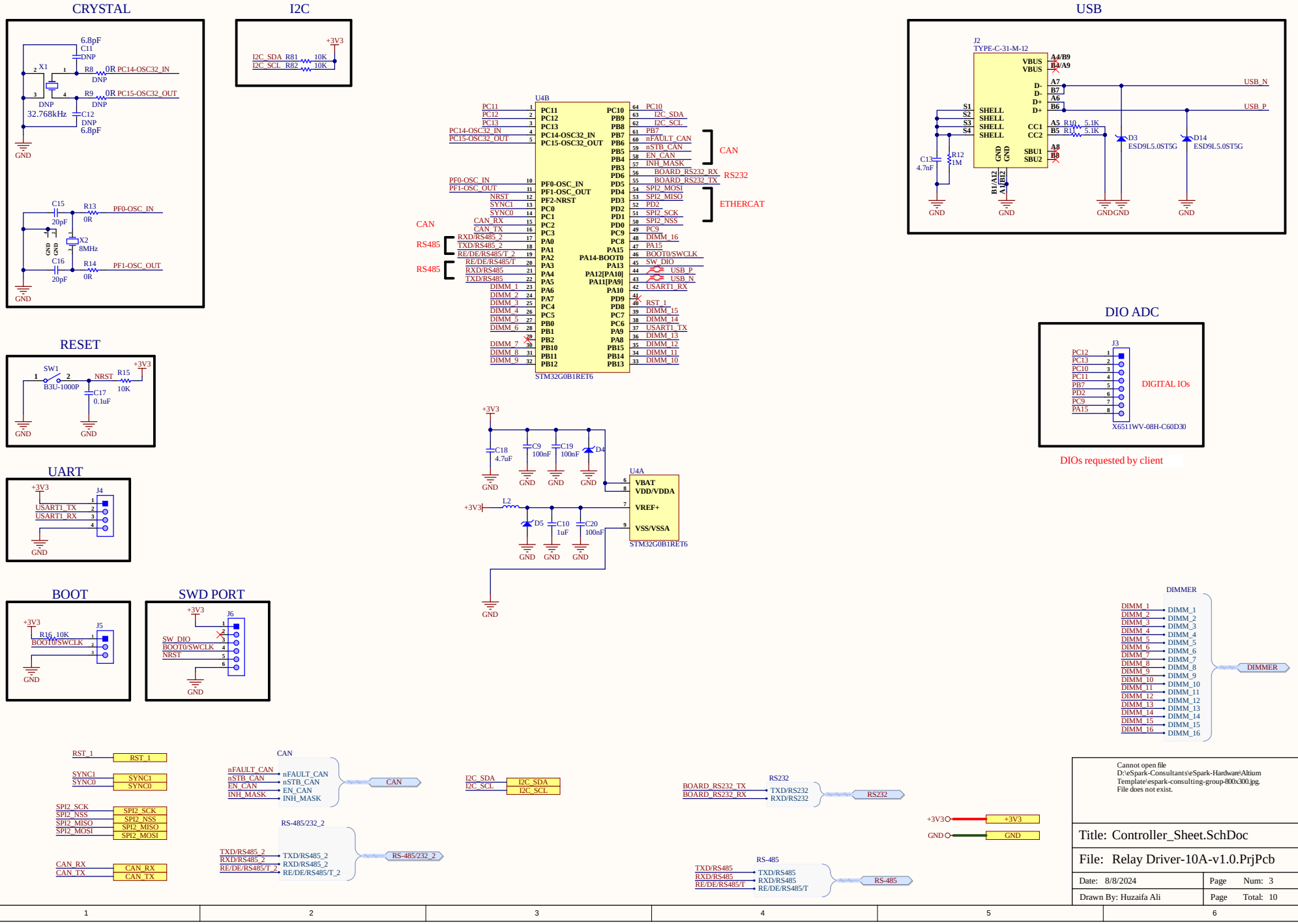
LEDs



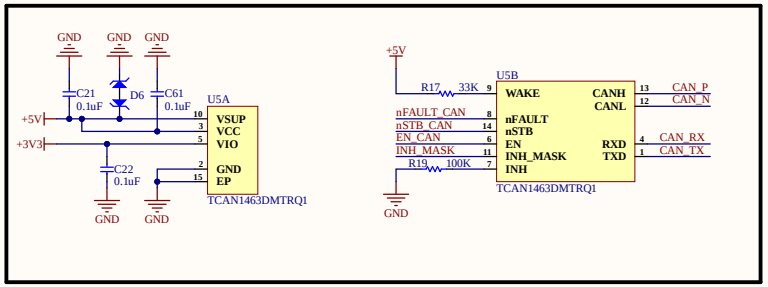


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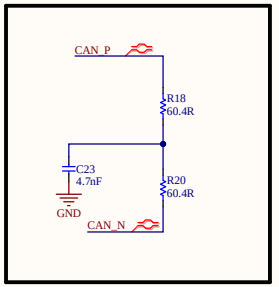
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File: Relay Driver-10A-v1.0.PrjPcb		
Date: 8/8/2024	Page	Num: 2
Drawn By: Huzaifa Ali	Page	Total: 10


CRYSTAL
6.8pF
C11
DNP
X1
R8 0R PC14-OSC32_IN
DNP
R9 0R PC15-OSC32_OUT
DNP
DNP
32.768kHz
C12
DNP
6.8pF
GND
C15
20pF
R13
0R
PF0-OSC_IN
X2
8MHz
C16
20pF
R14
0R
PF1-OSC_OUT
GND
I2C
+3V3
I2C_SDA R81 10K
I2C_SCL R82 10K
U4B
STM32G0B1RET6
PC11
PC12
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PF0-OSC_IN
PF1-OSC_OUT
NRST
SYNC1
SYNC0
CAN_RX
CAN_TX
RXD/RS485_2
TXD/RS485_2
RE/DE/RS485/T_2
RE/DE/RS485/T
RXD/RS485
TXD/RS485
DIMM_1
DIMM_2
DIMM_3
DIMM_4
DIMM_5
DIMM_6
DIMM_7
DIMM_8
DIMM_9
PC10
PB9
PB8
PB7
PB6
PB5
PB4
PB3
PD6
PD5
PD4
PD3
PD2
PD1
PD0
PC9
PC8
PA15
PA14-BOOT0
PA13
PA12[PA10]
PA11[PA9]
PA10
PD9
PD8
PC7
PC6
PA9
PA8
PB15
PB14
PB13
I2C_SDA
I2C_SCL
nFAULT_CAN
nSTB_CAN
EN_CAN
INH_MASK
BOARD_RS232_RX
BOARD_RS232_TX
SPI2_MOSI
SPI2_MISO
PD2
SPI2_SCK
SPI2_NSS
PC9
DIMM_16
PA15
BOOT0/SWCLK
SW_DIO
USB_P
USB_N
USART1_RX
RST_1
DIMM_15
DIMM_14
USART1_TX
DIMM_13
DIMM_12
DIMM_11
DIMM_10
CAN
RS485
RS485
RS232
ETHERCAT
USB
J2
TYPE-C-31-M-12
VBUS
CC1
CC2
SBU1
SBU2
D-
D+
SHELL
R10 5.1K
R11 5.1K
C13 4.7nF
R12 1M
D3 ESD9L5.0ST5G
D14 ESD9L5.0ST5G
RESET
SW1
B3U-1000P
NRST
R15 10K
C17 0.1uF
UART
J4
USART1_TX
USART1_RX
BOOT
J5
R16 10K
BOOT0/SWCLK
SWD PORT
J6
SW_DIO
BOOT0/SWCLK
NRST
+3V3
C18 4.7uF
C9 100nF
C19 100nF
D4
L2
D5
C10 1uF
C20 100nF
U4A
VBAT
VDD/VDDA
VREF+
VSS/VSSA
STM32G0B1RET6
DIO ADC
J3
PC12
PC13
PC10
PC11
PB7
PD2
PC9
PA15
DIGITAL IOs
X6511WV-08H-C60D30
DIOs requested by client
DIMMER
DIMM_1
DIMM_2
DIMM_3
DIMM_4
DIMM_5
DIMM_6
DIMM_7
DIMM_8
DIMM_9
DIMM_10
DIMM_11
DIMM_12
DIMM_13
DIMM_14
DIMM_15
DIMM_16
RST_1
SYNC1
SYNC0
SPI2_SCK
SPI2_NSS
SPI2_MISO
SPI2_MOSI
CAN_RX
CAN_TX
CAN
nFAULT_CAN
nSTB_CAN
EN_CAN
INH_MASK
I2C_SDA
I2C_SCL
RS-485/232_2
TXD/RS485_2
RXD/RS485_2
RE/DE/RS485/T_2
RS232
BOARD_RS232_TX
BOARD_RS232_RX
TXD/RS232
RXD/RS232
RS-485
TXD/RS485
RXD/RS485
RE/DE/RS485/T
GND
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Date: 8/8/2024
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Drawn By: Huzaifa Ali
Page Total: 10

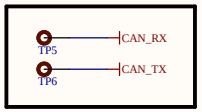
CAN BUS



CAN TERMINATION



TESTPOINTS





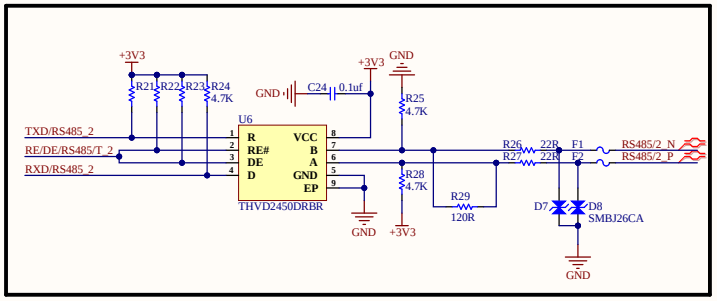




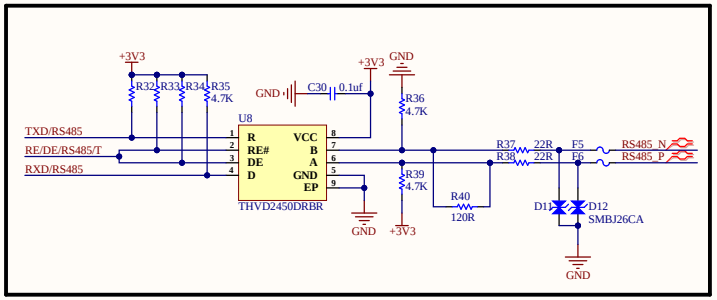
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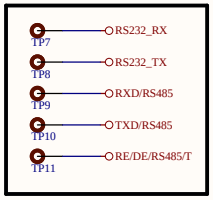
RS485_2 TO UART



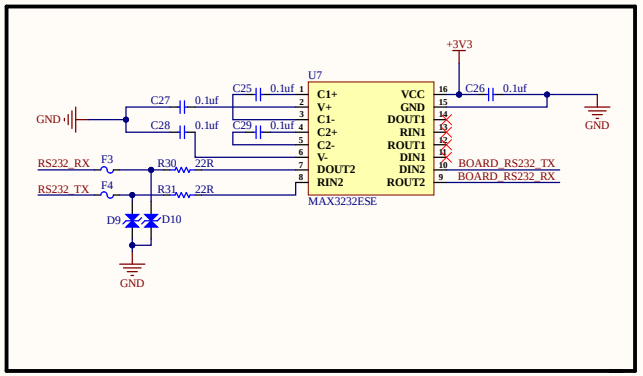
RS485 TO UART



TESTPOINTS



RS232 TO UART



RS485/2_N
RS485/2_P
RS485/2_N
RS485/2_P

RS485_N
RS485_P
RS485_N
RS485_P

RS232_RX
RS232_TX
RS232_RX
RS232_TX

RS-485/232_2
TXD/RS485_2
RXD/RS485_2
RE/DE/RS485/T_2
TXD/RS485_2
RXD/RS485_2
RE/DE/RS485/T_2
RS-485/232_2

RS232
BOARD_RS232_TX
BOARD_RS232_RX
TXD/RS232
RXD/RS232
RS232

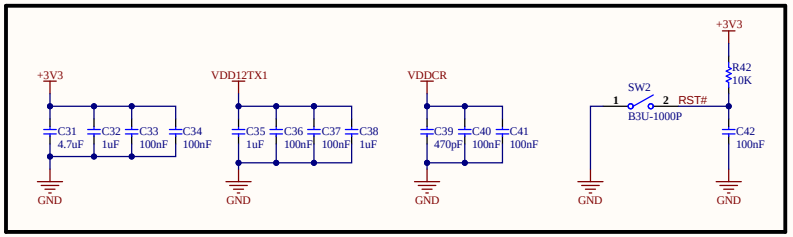
RS-485
TXD/RS485
RXD/RS485
RE/DE/RS485/T
TXD/RS485
RXD/RS485
RE/DE/RS485/T
RS-485

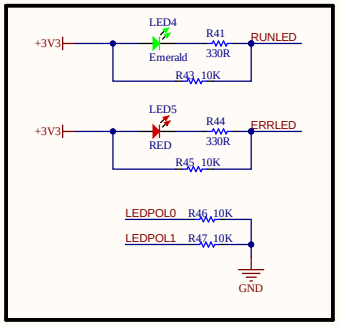
+3V3
+3V3

GND
GND

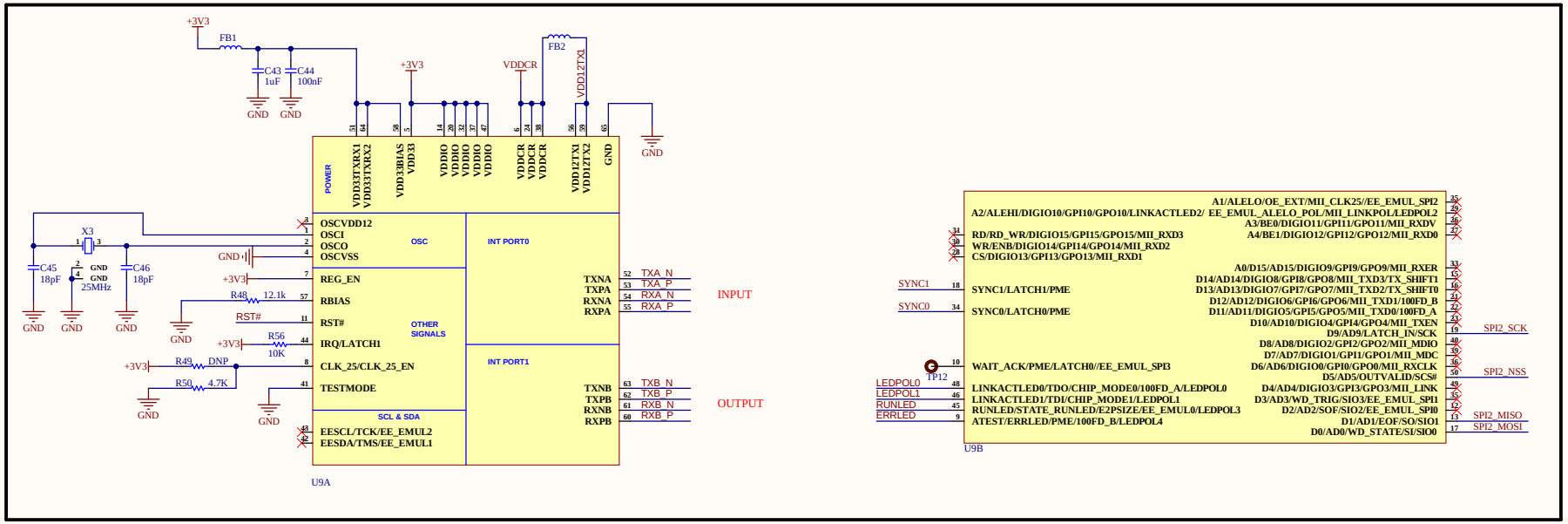
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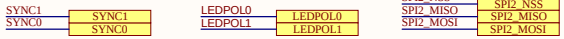
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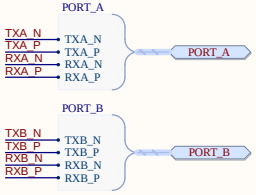




ETHERCAT





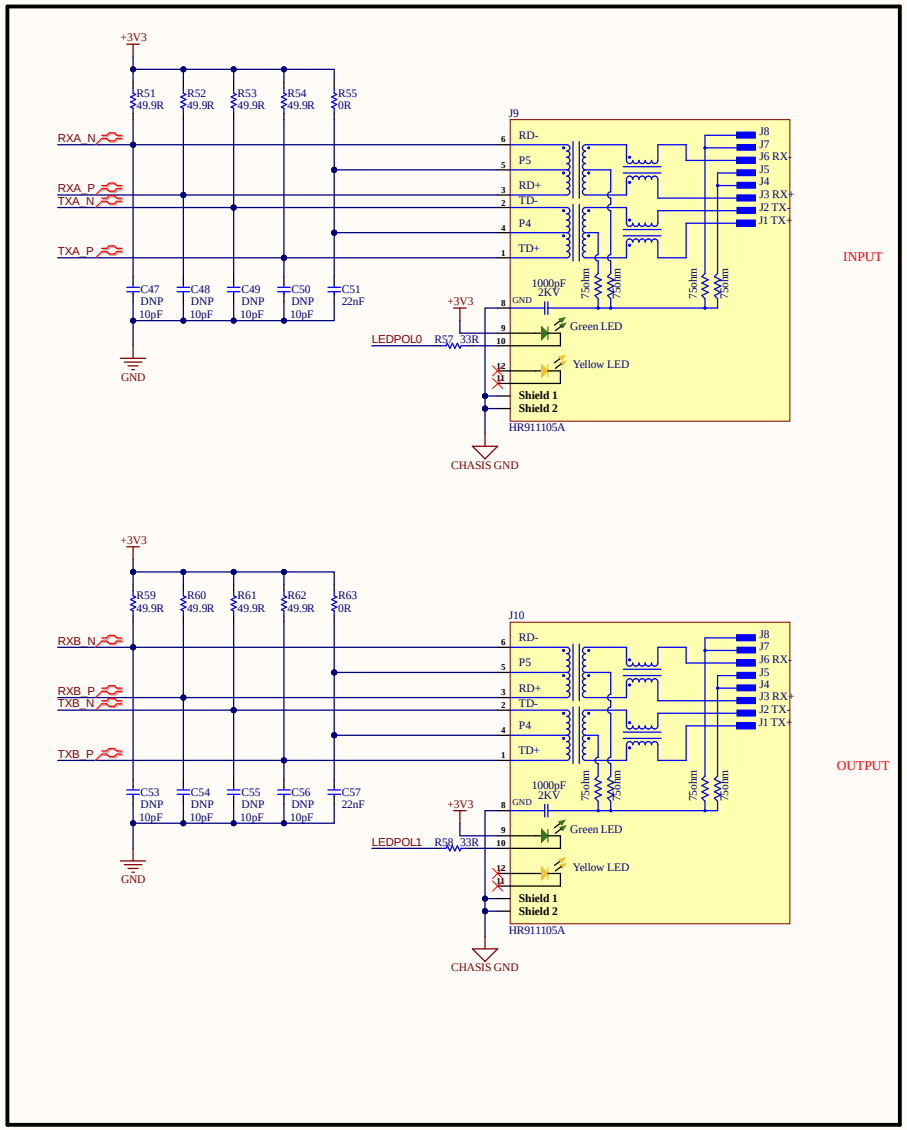




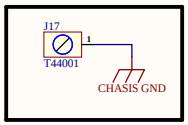
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File: Relay Driver-10A-v1.0.PrjPcb	
Date: 8/8/2024	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 10

RJ45



EARTH TERMINAL



LED1 (Green) = LINK/ACT
LED2 (Yellow) = SPEED

PORT_A
TXA_N
TXA_P
RXA_N
RXA_P
TXA_N
TXA_P
RXA_N
RXA_P
PORT_A

PORT_B
TXB_N
TXB_P
RXB_N
RXB_P
TXB_N
TXB_P
RXB_N
RXB_P
PORT_B

LEDPOL0
LEDPOL1
LEDPOL0
LEDPOL1

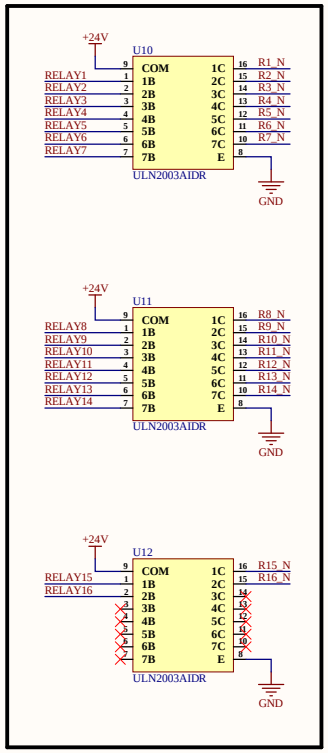
+3V3
+3V3
GND
GND

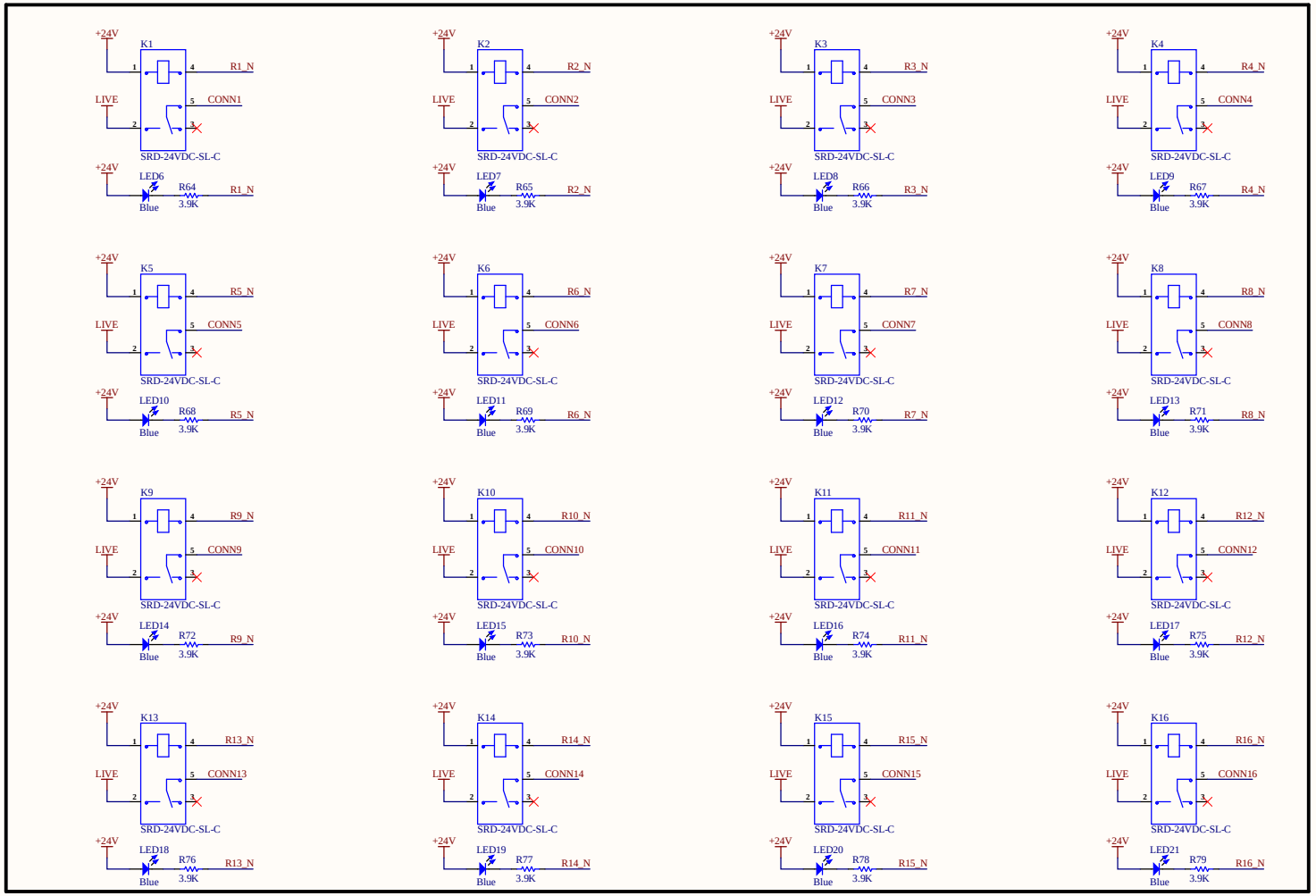
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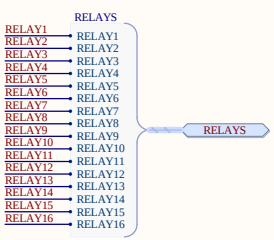
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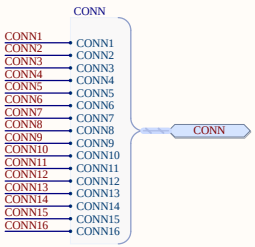
RELAYS 10A TRACES

RELAY DRIVER









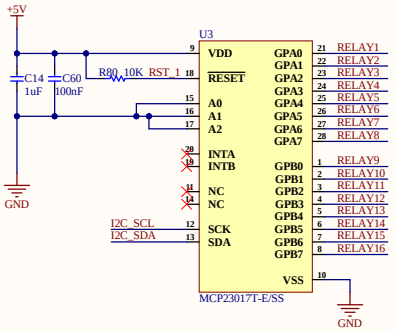


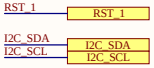


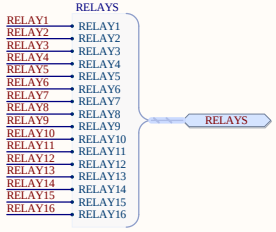
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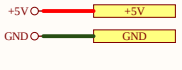
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GPIO EXPANDER





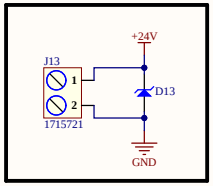




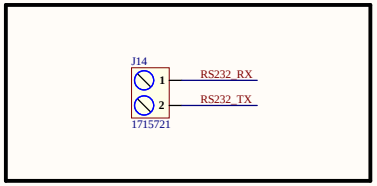
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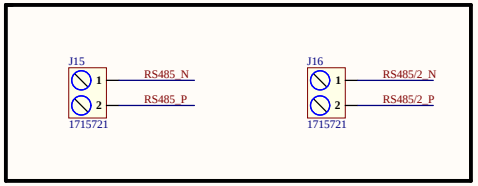
+24V OUTPUT



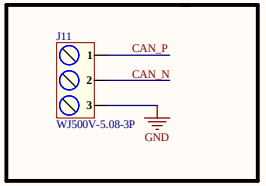
RS-232



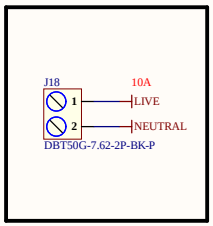
RS-485



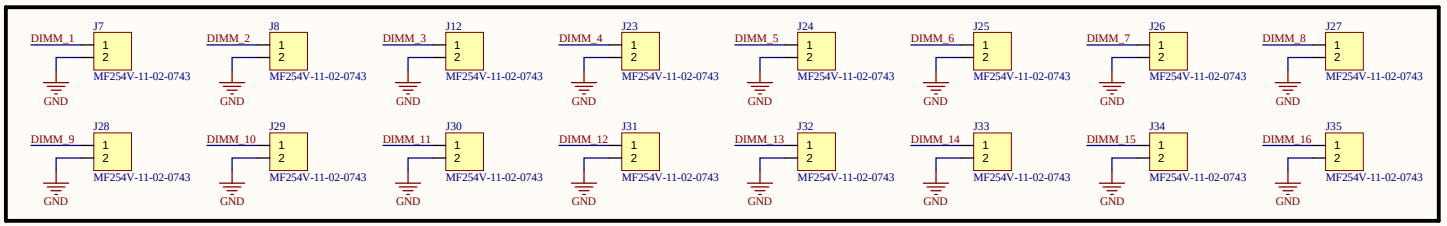
CAN



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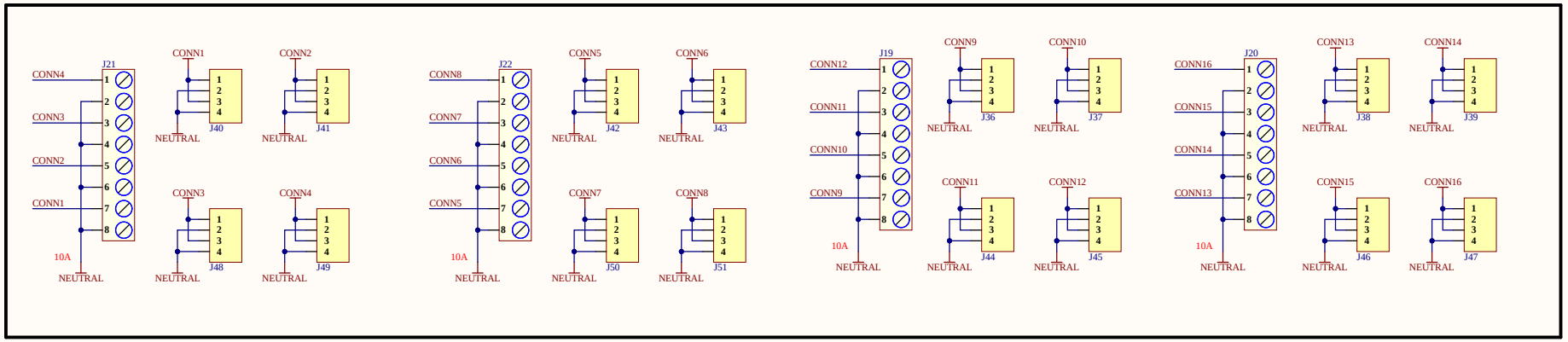


DIMMER CONNECTOR

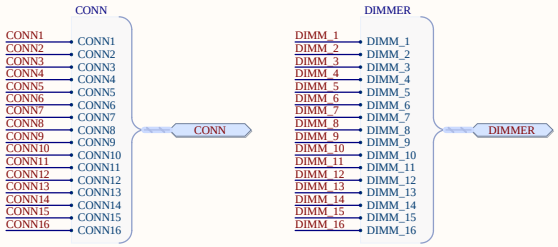


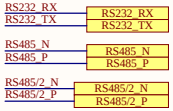
For Daughter Card PWM Signal

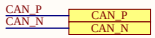
RELAY CONNECTION EVERY TRACE IS 10A



For Daughter Card Power 220v 10A











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File: Relay Driver-10A-v1.0.PrjPcb	
Date: 8/8/2024	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 10